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REVIEW ARTICLE

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A Competency Framework for AI Literacy: Variations by Different Learner Groups and an Implied Learning Pathway

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This study aims to develop a comprehensive competency framework for artificial intelligence (AI) literacy, delineating essential competencies and sub-competencies. This framework and its potential variations, tailored to different learner groups (by educational level and discipline), can serve as a crucial reference for designing and implementing AI curricula. However, the research on AI literacy by target learners is still in its infancy, and the findings of several existing studies provide inconsistent guidelines for educational practices. Following the 2020 PRISMA guidelines, we searched the Web of Science, Scopus, and ScienceDirect databases to identify relevant studies published between January 2012 and October 2024. The quality of the included studies was evaluated using QualSyst. A total of 29 studies were identified, and their research findings were synthesized. Results show that at the K-12 level, the required competencies include basic AI knowledge, device usage, and AI ethics. For higher education, the focus shifts to understanding data and algorithms, problem-solving, and career-related competencies. For general workforce, emphasis is placed on the interpretation and utilization of data and AI tools for specific careers, along with error detection and AI-based decision-making. This study connects the progression of specific learning objectives, which should be intensively addressed at each stage, to propose an AI literacy education pathway. We discuss the findings, potentials, and limitations of the derived competency framework for AI literacy, including its theoretical and practical implications and future research suggestions.

KEYWORDS

AI literacy, learner competency framework, learner group, pathway

Practitioner notes

What is already known about this topic

- AI literacy is becoming increasingly important as AI technologies are integrated into various aspects of life and work.
- Research on AI literacy competencies across diverse learner groups and disciplines remains fragmented and inconsistent to guide educational practices.
- Studies providing a coherent pathway for AI literacy development throughout educational and working life are lacking.

What this paper adds

- A comprehensive AI literacy competency framework consisting of 8 competencies and 18 sub-competencies.
- Variations in AI literacy competencies with tailored configuration and prioritization across different learner groups by school levels and disciplines.
- A proposed pathway for developing AI literacy from K-12 to higher education and workforce levels.

Implications for practice and policy

- The framework can guide the design and implementation of AI curricula tailored to different learner characteristics and needs.
- Education should shift focus from teaching how to use AI to fostering competencies for critical, strategic, responsible and ethical integration of AI.
- Policies are needed to support a systematic pathway for lifelong AI literacy development from K-12 education to workforce training.

INTRODUCTION

The integration of artificial intelligence (AI) into every facet of life continues at an ever-increasing pace, exerting significant impact on individuals and society as a whole. AI, which ranges from conversational agents such as chatbots to more advanced systems such as humanoid robots, is widely considered a transformative technology that will shape the future of every industry, including, but not limited to, computer-related fields and areas such as healthcare and education (Holmes et al., 2019; Luckin et al., 2016). For example, in healthcare, AI is used for diagnostic support, treatment planning, and drug discovery. In education, AI is used to personalize learning and improve student engagement and assessment (Luckin et al., 2016; Pearson, 2019). The impact of AI is anticipated to be both far-reaching and profound and will virtually change all human activities and endeavours.

As AI has the potential to revolutionize the way we work, live, and interact with one another by automating repetitive tasks, improving decision making, and facilitating communication, it is widely agreed that the job market and the competencies required of individuals will also change (Frey & Osborne, 2017). Frey and Osborne (2017) found that jobs highly susceptible to computerization include those in the transportation, logistics, and administrative sectors. A study by the World Economic Forum (2018) revealed that jobs requiring skills in areas such as complex problem solving, critical thinking, and creativity are expected to increase while jobs requiring skills in areas such as data entry and processing are expected to decline. Meanwhile, a study by the European Commission (2019) highlighted the projected increase in jobs requiring skills in areas such as communication, collaboration, and emotional intelligence.

The growing consensus is that the changes in the competencies required of individuals due to social changes caused by AI necessitate a corresponding change in education (Kirkwood & Price, 2019; Li et al., 2018; Martinez-Maldonado & Perez-Sanagustin, 2019). Kukulska-Hulme, Beirne, et al. (2020) proposed the concept of *education FOR AI* (preparing individuals to work with AI and develop AI systems) and *education WITH AI* (using AI to enhance teaching and learning), with the former encompassing a broader scope. Educational research, particularly into AI literacy, should provide critical and fundamental guidance for *education FOR AI*. The prevalence of AI demands the acquisition of new knowledge because individuals need a comprehensive understanding of the AI technologies and phenomena to adapt effectively to these changes (Yi, 2021). Literacy now entails more than just the ability to read and write, as traditionally referred to; it includes the ability to network and access a wider range of information (Long & Magerko, 2020). Efforts to increase literacy have been linked to political and social empowerment. Literacy has evolved from being a technology exclusive to experts to becoming an integral part of everyday life. Similar to the development of writing, AI has entered the realm of everyday life, becoming a goal of basic literacy.

As technology advances, research into various forms of literacy—*computer literacy*, *ICT (Information and Communication Technology) literacy*, and *digital literacy*—has evolved over the past several decades to address an increasingly broad range of abilities, reflecting rapid developments in technology. *Computer literacy* studies began in the 1960s and 1970s with the advent of personal computers and defined computer literacy as one's ability to use and understand basic computer hardware and software. Focusing primarily on the technical skills required to operate computers, it included an understanding of how to input data, navigate and use the operating system, and perform basic tasks such as creating and editing documents (Lee & Mestre, 1999; Reynolds, 2015; Warschauer, 2003). *ICT literacy* studies began in the 1980s and 1990s as the internet and other forms of communication technology became more widely available. These studies expanded the definition of computer literacy to include the use and understanding of various forms of communication technology, such as the internet, email, and mobile devices. They also deemed ICT literacy to include the ability to use these technologies and to access, manage, and create information (Livingstone, 2008; UNESCO IITE, 2018). *Digital literacy* studies began in the early 2000s as the use of digital technology continued to expand and the internet became a central part of everyday life. Digital literacy is considered an even broader concept than ICT literacy and includes not only the ability to use technology but also the ability to critically evaluate, create, and communicate information using digital media (Eshet, 2004). Recent studies on digital literacy suggest that, to respond more agilely and responsibly to the technological advancements of the AI era, it is necessary to expand the scope of digital literacy (Tiernan et al., 2023).

Research on AI literacy emerged in the early 2000s, initially overlapping with digital literacy studies, particularly in areas such as digital transformation, ethics, and responsible AI (European Commission, 2019). However, AI literacy, as it is understood today, has gained momentum more recently, particularly after 2020, reflecting the growing prevalence of AI technologies in everyday life (Miao et al., 2021). The concept evolved from focusing on basic AI concepts and technologies to encompassing a broader scope that includes understanding social and ethical implications, as well as using AI across various domains. Ng et al. (2021) define AI literacy as the ability to understand, use, monitor, and critically reflect on AI applications, emphasizing not only comprehension but also application and evaluation in diverse contexts. AI literacy now became to aim to equip individuals with skills for effective interaction with AI systems, informed decision-making, and understanding AI's broader societal implications, reflecting the increased prevalence and sophistication of AI technologies.

Given that AI literacy can be defined as a comprehensive set of competencies enabling individuals to critically evaluate AI technologies, effectively communicate and collaborate with AI, and utilize AI as a tool in various settings, such as online, at home, and in the workplace (Long & Magerko, 2020), research on the competency framework of AI literacy can serve as a crucial reference for designing and implementing AI curricula (Kong et al., 2023). Several efforts have attempted to list relevant competencies. Long and Magerko (2020) proposed 17 AI competencies, which include recognizing AI, understanding the concept of intelligence, identifying the steps involved in machine learning, critically interpreting AI outputs, data literacy, and adhering to ethical standards in AI usage. Almatrafi et al. (2024) identified AI literacy competencies as recognizing, knowing and understanding, using and applying, evaluating, creating, and navigating ethically. Both studies emphasize a comprehensive approach to AI literacy, focusing on understanding and utilizing AI. In contrast, Huang (2021) identified key AI competencies with a greater emphasis on cognitive and soft skills, categorizing them into knowledge, teamwork, and learning domains, further subdividing them into skill and cultural competence, teamwork and human-tool collaboration competence, and cognitive and self-learning competence. This difference in focus underscores the varied perspectives within AI literacy research, with some studies concentrating on the technical understanding and application of AI, while others emphasize the cognitive and interpersonal skills necessary for effective AI engagement. The findings from available studies remain inconsistent and lack a comprehensive framework that integrates these diverse perspectives.

Furthermore, while studies on AI competencies exist for specific learner groups, a comprehensive framework spanning the entire life stage continuum is lacking. Research has focused on various educational levels, including early childhood (Su et al., 2023), K-12 (Casal-Otero et al., 2023; Ng et al., 2023; Sanusi et al., 2022), and higher and adult education (Laupichler, Aster, et al., 2022; Ng et al., 2021), often yielding inconsistent results. For instance, Casal-Otero et al. (2023) emphasized understanding AI operations and using AI tools, while Sanusi et al. (2022) proposed key competencies for students with notable emphasis on team competence, including teamwork and human-tool collaboration. In contrast, Ng et al. (2023) proposed a broader framework that addresses affective, behavioural, cognitive, and ethical domains. Additionally, studies have explored AI competencies within specific disciplines such as engineering (Cantú-Ortiz et al., 2020), medicine (Karaca et al., 2021; Wartman & Combs, 2019), nursing (Swan, 2021), and communication (Dergaa et al., 2023). However, comparative analyses across multiple disciplines remain scarce. This fragmented approach highlights critical gaps: the need to systematically examine how AI competencies vary by educational level and discipline, and the absence of research that provides a comprehensive pathway for developing AI literacy competencies throughout one's educational and working life. This study aims to address these caveats and provide a comprehensive overview of composing competencies for an AI literacy framework with its variations by different learner groups and possible pathways towards becoming AI-literate. We have accordingly posed the following research questions:

1. What competencies and sub-competencies constitute an AI literacy framework?
2. How do these competencies vary by educational level and discipline?
3. What could be a possible pathway for equipping learners with competencies in AI literacy, starting from K-12 through college students and extending to working adults and the general public?

METHODS

This review was planned, conducted, and reported according to the updated guidelines for reporting systematic reviews of the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Page et al., 2021).

Search strategy

We searched Web of Science (1900–present), Scopus (1970–present), and ScienceDirect (1995–present) for records from January 2012 (the year when interest in AI from academia and industry started to increase) to October 2024. Using Boolean operators and wildcards, we grouped search terms into three categories: (1) AI (AI OR Artificial intelligence*), (2) education (education* OR learn* OR teach* OR instruct* OR school*), and (3) AI literacy (AI literacy OR artificial intelligence literacy OR competenc*OR capacit*). An experienced librarian verified the search strategies.

Inclusion and exclusion criteria

We included peer-reviewed journal publications from January 1, 2012 to October 1, 2024, written in English, focusing on AI literacy in education. Original research, including quantitative, qualitative, and review studies, was included; commentaries, letters, editorials, and book chapters were excluded. We included studies on K-12, higher education, vocational education, and general public education but excluded expert training for AI developers or programmers. We also excluded studies on using AI to train specific competencies required in certain areas, ie, education *with* AI (Kukulska-Hulme, Hammond, & Sharples, 2020, p. 11) (Table A1). Three authors independently and collaboratively screened titles, abstracts, and full texts of studies identified through our search. Disagreements were resolved through discussion until consensus was reached, with inclusion and exclusion criteria reviewed and refined during meetings for the first 100 items. We assessed inter-rater reliability using Fleiss Kappa, evaluating consistency by randomly selecting 30 articles at each stage. Fleiss's Kappa values were 0.75 for title screening, 0.78 for abstract screening, and 0.79 for full-text review, indicating excellent inter-rater reliability across all stages.

Data extraction

Three authors initially developed a comprehensive data extraction sheet through iterative testing and revision. This sheet was designed to capture key information from each study, including study characteristics, methodological details, participant information, definitions of AI literacy, identified AI competencies and sub-competencies, and target learner groups categorized by school level and discipline. The authors independently extracted data from the selected studies using this sheet. To ensure efficiency and thoroughness, they divided the studies among themselves. Each author meticulously reviewed their assigned studies, completing the data extraction sheet for each. Before proceeding with the full extraction process, the authors cross-checked a random sample of 20% of the studies ($n=5$) to verify the accuracy and consistency of the extracted data. Regular meetings were held throughout the extraction process to maintain consistency and resolve any discrepancies. During these meetings, they discussed any ambiguities in data interpretation and collectively addressed any issues that arose. In cases of disagreement, all three authors jointly reviewed the study in question and reached a consensus through discussion. The final data extraction sheet included only the data agreed upon by all authors after thorough review and discussion.

Data coding were conducted using a structured scheme that categorized studies based on three dimensions: research method, target learners, and AI literacy main/sub competencies. The research methods were classified as: 1.1 Quantitative (survey, scale development), 1.2 Qualitative (interview, observation, case study, curriculum design, scale development), 1.3 Mixed methods, and 1.4 Review (systematic review, narrative review). Target learners were categorized as: 2.1 K-12, 2.2 higher education (with specific disciplines), 2.3 teacher, 2.4 workforce, 2.5 general public and 2.6 organization. The AI literacy competencies were analysed, reconstructed and expanded based on the UNESCO Digital Literacy Global Framework, which includes: 3.1 Devices and software operations, 3.2 Information and data literacy, 3.3 Communication and collaboration, 3.4 Digital content creation, Safety, 3.5 Problem solving, and 3.6 Career-related competences.

Assessment of study quality

The quantitative and qualitative studies were assessed using QualSyst (Kmet et al., 2004), and review studies were assessed using the JBI Critical Appraisal Checklist for Text and Opinion (JBI, 2020). The Standard Quality Assessment Criteria comprise 14 items for quantitative studies and 10 items for qualitative studies. We followed the four-level quality standard based on the summary mean score percentage: >80% as strong, 70%–80% as good, 50%–70% as adequate, and <50% as low quality (Sim et al., 2016). One study, scoring 32%, was excluded. The average quality score of the 29 assessed studies was 77%, indicating strong and good quality. One study was assessed by the JBI Critical Appraisal Checklist for Text and Opinion, which has 6 items. We followed the 50% cutoff score (3 items) for inclusion (Hopia & Heikkilä, 2020). The study scored 83%; thus, no study was excluded. The three authors independently and collaboratively performed quality assessments and resolved substantial differences through discussions.

RESULTS

Trial flow

In total, 947 articles were identified from the three academic databases. After the removal of 253 duplicates, 694 potentially relevant articles were retained. Next, 225 articles remained eligible after 469 articles were eliminated based on their titles. In the abstract screening phase, 83 articles remained eligible after removing 142 articles. Then, 29 articles were included after 54 articles were removed during the full text screening phase. Figure 1 presents the exclusion process for article identification following the PRISMA guidelines.

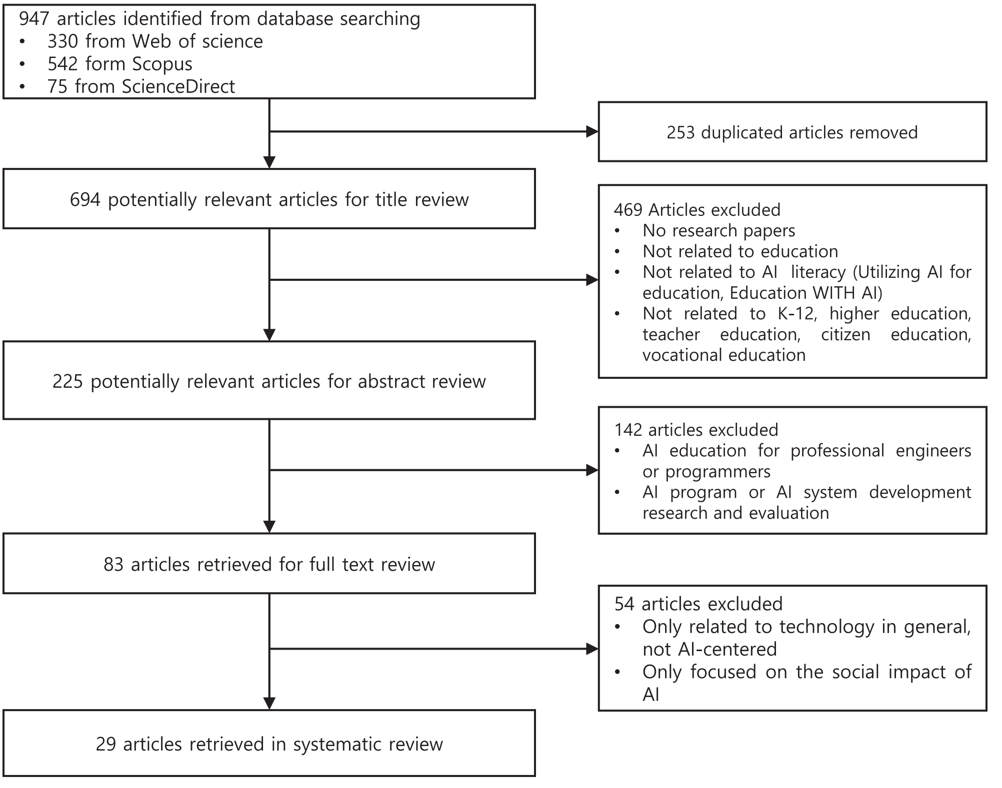


FIGURE 1 Trial flow for a systematic review of the learner competency for AI literacy.

Study characteristics of reviewed articles

Based on the analysis of the 29 articles, the most common type of study was quantitative (34%), followed by review studies (31%), mixed-method (24%), and qualitative studies (10%). Among the 17 studies with participants, 41% involved K-12 students, 24% college students, 18% teachers, 12% working adults, and 6% general public. The number of participants varied widely from 11 to 13,000. The most frequently reported research topics were (a) identifying key AI competencies ($n=17$, 59%), (b) developing curricula for AI literacy education ($n=6$, 21%), (c) teaching AI ethics ($n=4$, 14%), and (d) developing AI literacy scale ($n=2$, 7%). Detailed information on the reviewed studies is provided in Table A2.

RQ1. Competencies/sub-competencies comprising AI literacy

AI literacy competencies are defined by the knowledge of AI concepts and processes and the use of AI software, applications, and platforms ($n=25$, 86%); thinking and problem solving using AI technology ($n=14$, 48%); communicating and collaborating with AI, including human–tool and human–human collaborations ($n=13$, 45%); awareness of the opportunities and threats of AI and concern about AI ethics ($n=18$, 62%); and having confidence and self-efficacy in using AI technology ($n=3$, 10%). (For definitions of the AI literacy compared with digital literacy, see Table A3) We analysed the studies to identify the competencies and sub-competencies that constitute AI literacy.

Based on UNESCO Digital Literacy Global Framework, we analysed and reconstructed the AI literacy competencies from the 29 studies and obtained 8 competencies and 18 sub-competencies. (For a comparison between AI literacy and digital literacy competencies, see Table A4) The competencies are: (1) AI device and software/application/platform operations ($n=12$, 41%), (2) data and algorithm literacy ($n=23$, 79%), (3) problem solving ($n=14$, 50%), (4) communication and collaboration ($n=13$, 46%), (5) AI ethics ($n=18$, 62%), (6) career-related competences ($n=11$, 39%), (7) AI content creation ($n=7$, 25%), (7) affective competences ($n=3$, 11%). Most studies included multiple competencies and were counted in duplicate. The order of the competencies was determined by importance and difficulty. The final competency framework for AI literacy is presented in Table 1.

RQ2. Variations of competencies of different learner groups

Three studies included more than one target learner for AI literacy competencies; therefore, these studies were counted in duplicate, resulting in 36 targets derived from the 29 studies. 13 studies targeted K-12 students ($n=13$, 36%), 9 studies targeted college students ($n=9$, 25%), and 5 studies targeted workforce ($n=5$, 14%), 5 studies targeted teachers ($n=5$, 14%), 3 studies targeted general public ($n=3$, 8%), and one study targeted organizations ($n=1$, 3%). Categorizing the 29 studies by discipline, 2 studies were IT and engineering majors ($n=2$, 7%,), 4 studies were education ($n=4$, 14%), 1 study was communication ($n=1$, 3%), 5 studies were medicine ($n=5$, 17%), 1 study was nursing ($n=1$, 3%), and 10 studies did not identify discipline ($n=11$, 38%).

Tables 2 and 3 show the AI literacy competencies by learner group according to school level and discipline, respectively. The percentage represents the proportion of studies mentioning each competency to the total studies addressing AI literacy competencies at each education level.

The competencies of AI literacy required of K-12 learners were primarily centered on the *AI ethics* (26%), *data and algorithm literacy* (21%), and *device and software/application/platform operations* (21%). In higher education, the competencies with the highest priority were identified in the areas of *data and algorithm literacy* (25%), *problem solving* (18%), and *career-related competencies* (18%). The AI literacy demands for teachers were primarily centered on *data and algorithm literacy* (26%), *AI content creation* (21%) and *device and software operations* (16%). In the case of organizations, the most prominent AI literacy competencies were identified as *data and algorithm literacy*, *problem solving*, and *career-related competencies*, each accounting for 33% of the total. For general workforce, the most significant AI competency requirements were in the domains of *data and algorithm literacy* (21%) and *career-related competencies* (21%). Finally, in general public contexts, the most essential AI literacy competencies were *AI device and software/application/platform operations*, *data and algorithm literacy*, each accounting for 21% of the total.

Meanwhile, the competencies for AI literacy were drawn by discipline as well. In the field of technology, the highest priority was identified in the areas of *data and algorithm literacy* (24%) and *AI device and software/application/platform operations* (20%). In the field of engineering, the most prominent AI literacy competencies were identified as *data and algorithm literacy* and *problem solving* at 25% each. In education, the highest priorities were identified for *data and algorithm literacy* (24%), *AI content creation* (24%), *AI device and software/application/platform operations* (18%). In the field of communication, the essential competencies required were *AI device and software/application/platform operations*, *communication and collaboration, AI ethics*, and *career-related competencies*, each accounting for 25% of

TABLE 1 A final competency framework for AI literacy.

Competency frame	Description	Study #
1. AI device and software/application/platform operations	Build an environment for running AI processes and know how to use AI devices and software/applications/platforms	
1.1 Building an optimal environment for AI process	Build the optimal infrastructure and environment to utilize AI systems and select appropriate AI software/applications/platforms	4, 5, 10, 11, 12
1.2 Knowing how to operate AI software/application/platform	Have skills to operate AI software/application/platforms appropriately	4, 5, 6, 8, 10, 11, 12, 13, 14, 16, 25
2. Data and algorithm literacy	Understand the basic use of data and information, recognize how AI/algorithms work, and apply them	
2.1 Understanding data and information	Have statistical knowledge/skills and know how to search, process, and manage data	9, 11, 12, 13, 14, 18, 19, 21, 23, 24, 25, 26, 27
2.2 Knowing and applying knowledge about AI/algorithms	Know how AI/algorithms work, understand how they select and present information and data, and apply them in using AI software/applications/platforms	1, 3, 8, 10, 11, 12, 13, 14, 16, 18, 19, 20, 21, 22, 23, 24, 26, 27
3. Problem solving	Know what the problems are, select appropriate AI software, and make the best decisions through problem-solving processes based on AI technology	
3.1 Identifying and defining problems that should be solved using AI technology	Identify and define real-life problems and analyse, practice, and question them using AI technology	13, 18, 19, 20, 21, 25
3.2 Applying problem-solving processes using AI technology	Collect and process data, apply them in AI technology, and interpret the results	3, 6, 7, 10, 13, 14, 15, 18, 19, 21
3.3 Data-based decision making	Make appropriate decisions based on the results derived from AI technology	9, 13, 14, 16, 18, 19, 20, 21, 23
4. Communication and collaboration	Use AI technology to promote human-to-human and human-to-tool communication and collaboration skills	
4.1 Human-to-tool (AI) collaboration through AI technology	Find and use appropriate AI technologies to promote human-to-tool (AI) collaboration	4, 8, 9, 10, 12, 14, 18, 22, 24
4.2 Human-to-human communication with AI technology	Leverage AI technology to foster human-to-human communication and develop leadership and teamwork skills	4, 8, 12, 18, 20
5. AI ethics	Increase awareness of opportunities and threats of AI while considering AI ethics and social issues	
5.1 Being aware of the opportunities and threats of AI	Increase awareness of the strengths, weaknesses, opportunities, and threats of AI to generate ideas and know their impact on society	2, 4, 5, 13, 18, 19, 21, 22, 24, 26
5.2 Knowing and being concerned about AI ethics	Understand and consider the ethical issues involved in AI and follow ethical norms and regulations	4, 9, 12, 13, 14, 16, 19, 21, 22, 25, 26
6. Career-related competences	Operate specialized AI technologies and understand, analyse, and evaluate specialized data, information, and AI-based content for a particular field	

TABLE 1 (Continued)

Competency frame	Description	Study #
6.1 Operating specialized AI technologies for a particular field	Identify and use specialized AI software/applications/platforms for a particular field	3, 4, 9, 13, 14, 20, 22, 25, 26, 27
6.2 Interpreting and manipulating data, information, and AI content for a particular field	Understand, analyse, and evaluate specialized data, information, and AI-based content for a particular field in AI systems	9, 13, 14, 20, 23, 25, 26
6.3 Verifying AI results for accuracy and making rational decisions based on validated data	Recognize the potential for incorrect outcomes, verify AI-generated results for accuracy, and make rational decisions based on validated data	9, 13, 14, 26, 27
7. AI content creation	Use AI-based software to create content or run programming tools (Scratch, Python, etc.) to develop creative outcomes	
7.1 Creating AI content using AI tools	Create various outputs of AI software/applications (eg, images, texts) with creative thinking	7, 16, 18, 24
7.2 Programming	Develop purposeful and relevant AI content using programming tools (eg, Scratch, Python)	3, 6, 7, 10, 24
8. Affective competences	Have confidence and self-efficacy to develop interest and motivation in AI technologies, along with the ability for self-reflection in learning AI	
8.1 Confidence and self-efficacy in using AI	Having confidence and self-efficacy to promote interest and motivation in using AI technologies	5, 18, 26
8.2 Self-reflective mindset towards AI	Having a self-reflective mindset to continuously assess one's understanding of AI, identify one's level of AI literacy, and recognize areas that require further learning	5, 26

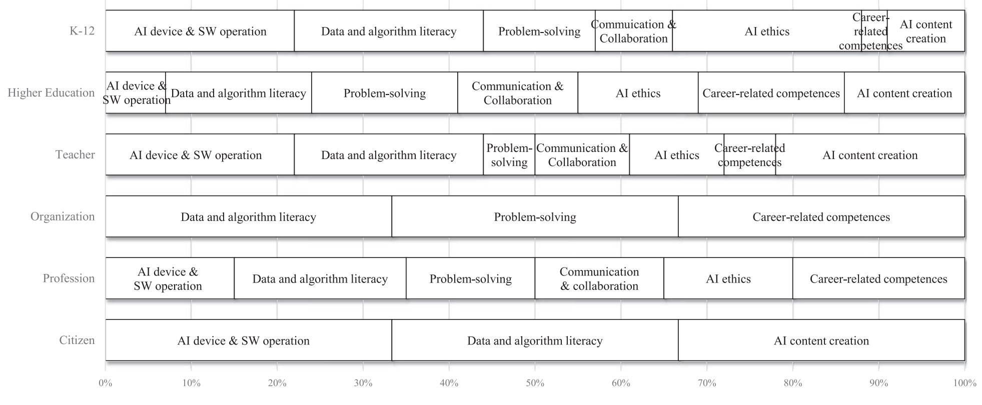
the total. In medicine, the most significant AI literacy requirements were in the domains of *data and algorithm literacy* and *career-related competencies*, each accounting for 22% of the total. In nursing, the most essential AI literacy competencies were identified as *data and algorithm literacy*, *problem solving*, *communication and collaboration*, and *career-related competencies*, each accounting for 25% of the total.

RQ3. A possible pathway for equipping learners with competencies for AI literacy

Based on the competencies required for K-12 students, college students, and general workforce depicted in Table 2, Figure 2 illustrates a possible pathway for equipping learners with the competencies for AI literacy that progresses from the K-12 level to the higher education and general workforce levels. In K-12 education, AI literacy focuses on acquiring the fundamental concepts and applications of AI. Furthermore, emphasis is placed on fostering awareness and understanding of the moral and social implications of AI, including the issues related to privacy and transparency. At the higher education level, AI literacy competencies tend to be more advanced and specialized. Learners at this stage are expected to possess the ability to solve problems by utilizing their foundational knowledge of AI and their proficiency in data and algorithms. These competencies are particularly relevant to career-related competencies that are of paramount importance in shaping career paths after graduation. Thus, the focus of AI education at the general workforce level is on developing career-related competencies, with particular emphasis on the ability to analyse information and data generated within a specific field.

TABLE 2 AI literacy competencies by learner group (school level).

	Competency 1	Competency 2	Competency 3	Competency 4	Competency 5	Competency 6	Competency 7	Competency 8
	AI Device and Software/ Application/ Platform Operations	Data and Algorithm Literacy	Problem Solving	Communication and Collaboration	AI Ethics	Career-related Competences	AI Content Creation	Affective Competences
K–12	21%	21%	7%	10%	26%	2%	7%	7%
Higher Education	6%	21%	18%	12%	15%	18%	6%	3%
Teachers	16%	26%	5%	11%	11%	5%	21%	5%
Organizations	0%	33%	33%	0%	0%	33%	0%	0%
Vocational Education	17%	21%	13%	8%	17%	21%	0%	4%
General Public	21%	21%	0%	14%	14%	0%	14%	14%



DISCUSSION

This study examined the AI literacy framework, investigated competency variations across educational stages and fields, and explored a pathway for equipping learners with AI literacy from K-12 to working life. Findings reveal distinct competency requirements ranging from basic AI knowledge and ethics at K-12 to advanced data analysis and problem-solving in higher education and workforce settings. Theoretical and practical implications, potentials, and limitations are discussed with future research suggestions.

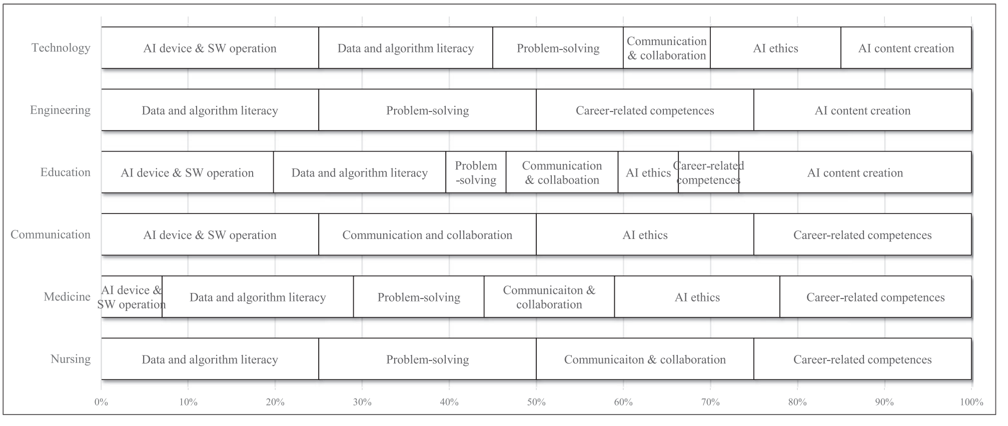
Competencies/sub-competencies of AI literacy

The main competencies for AI literacy include (1) AI device, software, application, and platform knowledge; (2) basic understanding and application of AI/data algorithms; (3) problem solving using AI software; (4) communication and collaboration in AI contexts; (5) AI ethics and social issues awareness; (6) career-specific AI competencies; (7) AI-based content creation; and (8) Affective competences in using AI.

AI literacy can be compared to computer literacy in the 1980s (eg, Papert, 1980) and digital literacy in the 1990s (eg, Gilster, 1997), which includes computer, ICT, information, and media literacy. This comparison shows how each literacy type aligns with

TABLE 3 AI literacy competencies by learner groups (discipline).

	Competency 1	Competency 2	Competency 3	Competency 4	Competency 5	Competency 6	Competency 7	Competency 8
	AI Device and Software/ Application/ Platform Operations	Data and Algorithm Literacy	Problem Solving	Communication and Collaboration	AI Ethics	Career-related Competences	AI Content Creation	Affective Competences
Technology	20%	24%	10%	15%	15%	2%	7%	7%
Engineering	0%	25%	25%	0%	0%	25%	25%	0%
Education	18%	24%	6%	12%	6%	6%	24%	6%
Communication	25%	0%	0%	25%	25%	25%	0%	0%
Medicine	11%	22%	19%	7%	19%	22%	0%	0%
Nursing	0%	25%	25%	25%	0%	25%	0%	0%



technological advancements and societal needs, demonstrating evolving skills necessary to navigate changing digital landscapes.

The transition from computer to digital literacy reflects the emergence of AI literacy. Computer literacy focused on understanding and using computer hardware and software, including operating software, basic computer abilities, and elementary problem-solving. As technology and social contexts changed, digital literacy expanded these concepts, defined by UNESCO (Law et al., 2018) as "the ability to access, manage, understand, integrate, communicate, evaluate and create information safely and appropriately through digital technologies for employment, decent jobs and entrepreneurship." (p. 6). This shift from computer to digital literacy signifies a move beyond technical proficiency to encompass skills and understanding necessary for effective functioning, participation, and responsibility in the digital world.

The advent of AI literacy extends this trend further, demanding a deeper understanding of the complexities and ethical considerations of AI technologies. AI literacy introduces a set of interesting capabilities (Tables A3 and A4).

First, understanding and applying knowledge about data and algorithms (2.2) is crucial as it enables individuals to discern how AI systems make decisions and how they can be applied in various contexts. This capability is complemented by data-based decision making (3.3), where leveraging vast amounts of data for rational and insightful conclusions becomes the norm, emphasizing the importance of critical thinking and analytical skills. Collaboration

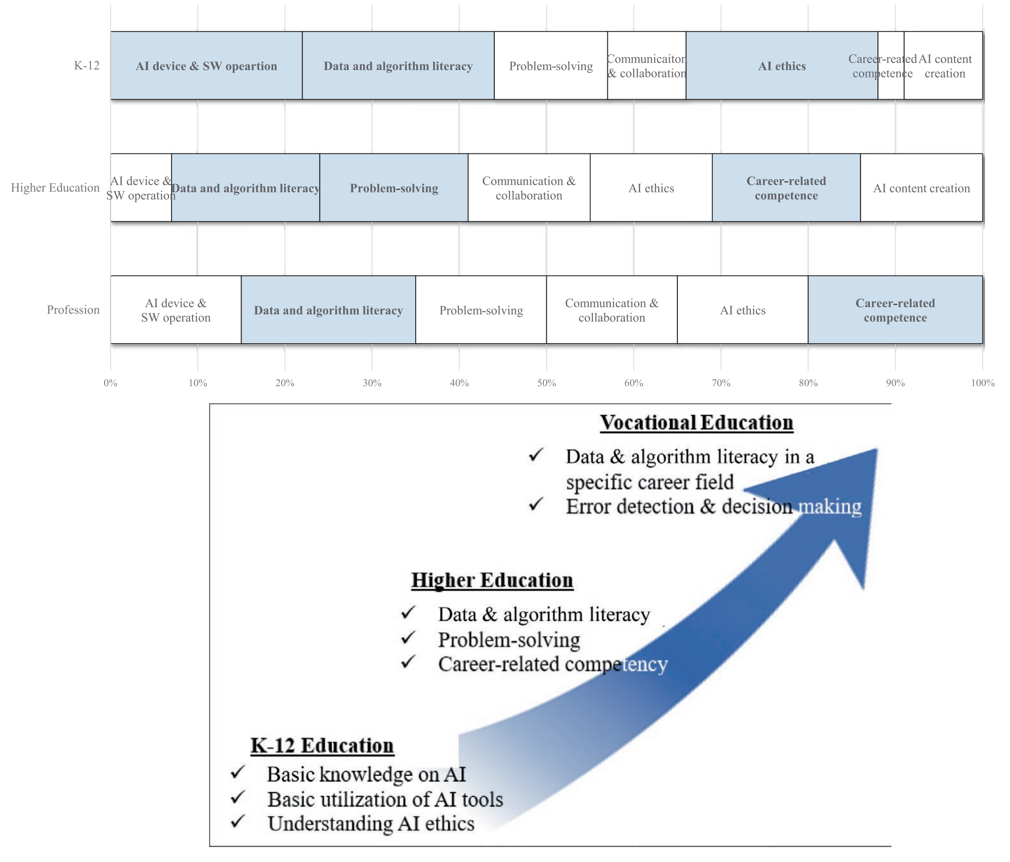


FIGURE 2 A possible pathway for educating competencies for AI literacy.

with AI technologies further enriches this literacy as it involves human-to-tool interaction (4.1), optimizing how we utilize AI to augment our capabilities. Equally critical is mastering human-to-human communication in an AI-mediated environment (4.2) and recognizing how AI can transform traditional communication channels and practices. Additionally, awareness of the dual-edged nature of AI, its opportunities and threats, rounds out this literacy, is necessary to maintain balanced perspective on the ethical, social, and professional impacts of AI. In the vocational context, this capability necessitates the ability to recognize the potential for incorrect outcomes, verify AI-generated results for accuracy, and make rational decisions based on validated data (6.3). The increasing influence of technology on society is evident in the transition from computer literacy to digital literacy and, subsequently, to AI literacy, aligning with Marshall McLuhan's media theory, which argues that technology is more than a tool; it functions as an environment that profoundly shapes society and culture, moulding our perception and experience of the world (McLuhan, 1964). Finally, cultivating affective competencies, including confidence and self-efficacy, is essential in the acquisition and application of AI knowledge (8.1). As learners build confidence in interacting with AI technologies, their interest and motivation to pursue academic and professional opportunities in the AI field are enhanced (Ng et al., 2023). This emotional shift serves as a key driver in developing AI-based problem-solving skills and career-relevant competencies.

The final AI literacy framework elucidates distinct differences when juxtaposed with similar studies, notably those of Ng et al. (2021, 2023), Long and Magerko (2020), and UNESCO (Cukurova & Miao, 2024) as depicted in Table A5. A salient distinction lies in the absence of career-related competencies. While the omission in Long and Magerko (2020) and Ng et al. (2023) could be attributed to its focus on K-12 students, the exclusion from Ng et al.'s (2021) study, which also addresses higher education, merits attention. Although it is challenging for higher education to fully meet the vocational needs of all students, the inclusion of career-related competencies, especially competency 6.3 (*Verifying AI results for accuracy and making rational decisions based on validated data*), which is crucial and not limited to specific professions—seems appropriate, aligning with societal demands. This notion is supported by other studies included in this analysis, which advocate for incorporating such competencies from the outset of higher education. The latest AI Competency Framework for Teachers, released by UNESCO in 2024, partially addresses these needs. The *AI for professional development* component emphasizes the capacity of teachers to adapt AI tools to support student professional growth and foster effective AI use. However, this component primarily focuses on the role of teachers in facilitating personalized learning and strategies within broader contexts of lifelong and collaborative organizational learning. As such, it diverges from the career-specific competencies outlined in this study, which emphasize skills for analysing, evaluating, and making informed decisions using AI technologies in specialized professional settings. In contrast to the four studies that only partially incorporate competencies or presented them within specific contexts, this study adopts a more comprehensive approach to amalgamating AI competencies and delineates how their composition and prioritization should vary according to the educational level and major of learner groups.

Competency variations across different learner groups

This study sets itself apart by delineating AI literacy competency requirements tailored to diverse educational levels and specific disciplines. It offers a unique contribution to the field by addressing the nuanced needs of different learner groups while considering the developmental stages and professional characteristics of each group. AI literacy education is not a one-size-fits-all model but rather a nuanced, multidimensional construct that caters to the specific needs and capabilities of different learner groups.

AI device and software/application/platform operations

This competency is generally emphasized at the K-12 and general public levels and in the field of education. Specifically, K-12 students need to develop a fundamental understanding of AI, along with the use of various AI devices, software, and applications, thereby laying a strong foundation for future AI education. It naturally becomes essential for education majors, who will be teaching these students, to possess the competency (Chiu & Chai, 2020; Kong & Abelson, 2019). For general public, possessing basic AI skills is becoming an essential competency in the AI era, similar to the daily use of cell phones and notebooks by ordinary individuals.

Data and algorithm literacy

This competency, highlighted as the most important across all learner levels, underscores the need for a deeper understanding of how AI and algorithms work. Dogruel et al. (2022) define algorithmic literacy as “the ability to understand algorithms, critically evaluate them, and have appropriate skills to cope with algorithmic systems” (p. 5). This competency is significant in higher education, general workforce, and organizational levels, where problem-solving and career-related competencies are required. Tuomi (2018) noted that while vocational education focuses on domain-specific knowledge, the demand for competencies like problem solving, critical thinking, and decision making is growing. This aligns with our findings on required competencies in medicine, healthcare, nursing, and engineering. Understanding AI models and their outputs is vital for model interpretability, enabling users to trust AI recommendations, explain decisions, and identify biases. For example, generative AI, optimized for pattern recognition and natural language processing, has limitations in traditional quantitative analysis. Choosing the right AI model for a specific task is vital for effectively using AI in problem solving, indicating the need for comprehensive educational strategies.

AI ethics

This competency is particularly essential within the K-12 group, reflecting the importance of students not only using AI effectively but also grasping its ethical and social implications. The OECD and UNICEF have emphasized the significance of AI ethics, including transparency, fairness, and accountability (Ng et al., 2021; UNICEF, 2019). Given that individuals aged 12–25 years are in the conventional moral period and are developing an understanding of social laws and moral norms (Kohlberg, 1984), K-12 learners must be educated about using AI responsibly and ethically and be equipped with appropriate values and perspectives (Hutchins et al., 2020). Among various disciplines, communication necessitates profound competency in AI ethics. The ability to discern accurate information from falsehoods and generate trustworthy content is imperative (Alser & Waisberg, 2023; Jarrah et al., 2023), implying a need for appropriate workforce training in this area.

Communication and collaboration

This competency was identified as the most necessary across disciplines, primarily because of the need to harness AI tools for specific tasks and to engage in discussions and collaborations (Swan, 2021). This requirement is most pronounced in nursing and communication, underscoring the need for proficiency in human–tool interactivity. This competency includes *human-to-tool collaboration through AI*, such as that in the use of AI-driven medical applications for patient management or generative AI for journalistic endeavours (Cantú-Ortiz et al., 2020; Dergaa et al., 2023); and *human-to-human collaboration through AI*, such as that in engagements in team-based discussions to adjust changing roles and workflows using AI-based tools (Russell et al., 2023).

AI content creation

This competency is predominantly observed among educators and engineering majors. In engineering education, efforts are increasingly focused on nurturing creativity among prospective engineers, underpinned by research advocating for creative outputs through programming language activities, which are becoming part of the curriculum for K-12 and engineering students (Hutchins et al., 2020; Lin et al., 2019). In parallel, AI content creation competency is essential for teachers designing and implementing AI-incorporated curricula. From the perspective of the Technological Pedagogical Content Knowledge (TPACK) framework, which outlines key knowledge for educators to effectively integrate technology into teaching, AI content creation competency engages with its three core components: technological knowledge (TK), pedagogical knowledge (PK), and content knowledge (CK) (Kim et al., 2021). This competency encompasses understanding and applying AI technologies (TK), integrating these technologies into teaching methodologies (PK), and applying AI within specific subject matter contexts (CK). AI content creation competency is essential for educators to prepare students for a future where technology and education intersect, ensuring teaching strategies are technologically advanced, pedagogically sound, and content-relevant, fostering an educational environment responsive to the needs of the AI era.

A possible pathway for equipping learners with competencies for AI literacy

This study presents an AI literacy competency framework and a developmental pathway for AI education across educational levels and life stages. It emphasizes the importance of implementing a long-term, strategic approach to provide age-appropriate AI education to ensure that individuals acquire the necessary knowledge and skills to navigate an AI era. The findings of this study, particularly regarding which competencies are crucial at different stages, warrant discussion in light of human cognitive and moral development theories (eg, Piaget (1952), Kohlberg (1984)), as well as in comparison with previous AI literacy research.

K-12 education, according to our findings, focuses on developing fundamental AI knowledge, including basic programming concepts and AI systems. More importantly, K-12 education prioritizes AI ethics to address ethical issues and dilemmas in AI development and deployment, promoting responsible AI usage (Kong et al., 2023; Zhang et al., 2023). These competencies correspond to Piaget's (1952) concrete operational stage and Kohlberg's (1984) conventional level of moral reasoning. This alignment is evident in the emphasis on developing logical thinking about tangible AI concepts and internalizing ethical norms for AI use, as highlighted by Kong et al. (2023) and Zhang et al. (2023). These findings also align with the three core dimensions of AI literacy competencies—AI concepts, AI applications, and AI ethics—presented in the studies by Long and Magerko (2020) and Casal-Otero et al. (2023). Curriculum designers should align AI education with students' cognitive and moral developmental stages, introducing AI concepts through tangible examples for younger students and progressing to more abstract concepts and complex ethical scenarios as students grow. AI ethics education should evolve from simple rules-based approaches to nuanced discussions of ethical dilemmas, reflecting students' moral reasoning development (Holmes et al., 2019). In higher education, the focus shifts to developing in-depth AI knowledge, including data and algorithm literacy and AI-based problem-solving skills. This shift aligns with Piaget's formal operational stage and the development of abstract reasoning abilities. Students engage in tasks such as developing basic AI algorithms to address specific challenges, which deepens their understanding of AI and its practical applications (Schmohl et al., 2022; Yadav et al., 2012). This progression reflects the maturation of higher-order thinking skills, specifically computational thinking (Wing, 2006) and algorithmic thinking (Futschek, 2006), enabling students to solve complex problems using AI. The emphasis on problem-solving and AI content creation in our study underscores the importance of these advanced cognitive skills, which align with the *Use and Apply* and *Evaluate and Create* competencies proposed by Ng et al. (2021) in their AI literacy framework. To foster these competencies, Laupichler, Hadizadeh, et al. (2022) advocate for practice-oriented curricula that cultivate higher-order thinking skills. Furthermore, collaborations between educational institutions and industry partners can enhance this approach, bridging theoretical knowledge with real-world applications (Cantú-Ortiz et al., 2020) and preparing students for the cognitive demands of AI-driven workplaces.

In workforce training, the focus is on developing field-specific AI literacy reflecting the diverse and specialized nature of working environments. Research on AI literacy at the workforce education level identifies competencies such as determining the appropriateness of technology for a given situation, using AI programs to analyse data and interpret results, and explaining these findings to colleagues or patients (Laupichler, Hadizadeh, et al., 2022; Malerbi et al., 2023; McCoy et al., 2020). This study identified these competencies as *career-related competences*, which involve the ability to utilize specialized AI technologies and understand, analyse, and evaluate data, information, and AI-based content specific to particular fields. There is a need for educational programs that strengthen these career-related competences to ensure the effective use of AI in the workplace. Workforce AI literacy education should adopt andragogical approaches that emphasize self-directed, experience-based learning (Knowles, 1984) and incorporate cycles of concrete experience, reflection, conceptualization, and active experimentation with AI technologies in professional contexts (Kolb, 1984). While it is crucial to tailor AI literacy education to specific career needs, it is equally important to maintain a focus on developing transferable, interdisciplinary AI competencies that support boundaryless career trajectories and enable professionals to adapt to the evolving AI landscape across various fields (Arthur & Rousseau, 1996).

Affective competencies are critical for the general public as they emphasize the importance of cultivating traits such as confidence and self-efficacy, enabling individuals to effectively acquire knowledge and adapt within the rapidly evolving landscape of AI technology. These competencies, inherently linked to emotional factors, shape individuals' attitudes towards AI, which in turn influence their interest, intention, and willingness to engage with AI in learning contexts (Compeau & Higgins, 1995; Davis, Bagozzi, & Warshaw, 1989; Ferdousi, 2019; Ng et al., 2023). Thus, it is imperative to design pedagogical strategies and curricula that embed affective competencies within adult and lifelong education, fostering sustained motivation, engagement, and AI literacy among the public.

This study also highlights the importance of maintaining general AI literacy for adults and seniors. As AI technologies increasingly integrate into everyday life, it is essential for individuals of all ages to grasp the basics of AI and its implications. Lifelong learning initiatives, community workshops, and online resources can offer accessible and engaging AI education tailored to adult learners. This study underscores the importance of specialized AI literacy education tailored to different disciplines, requiring accurate analysis of potential impacts of AI and discipline-specific curricula development. It demonstrates that AI literacy comprises concrete competencies that evolve through life stages, paralleling educational adaptation to human development, cognitive growth, and vocational growth. The study is meaningful because it found that AI literacy can be embodied in each core and sub-competency, forming an organically connected flow rather than independent concepts.

In a society intertwined with AI, understanding, interpreting, and utilizing AI will be a fundamental life competency. The AI competency framework shows that effective AI utilization necessitates general competencies, such as critical thinking, effective communication, collaboration with people and AI tools, ethical judgement and affective competency such as self-efficacy in the use of AI. This underscores the critical role of "human consciousness" as the cornerstone of these competencies. While AI can process data, it lacks the ability to make decisions based on critical considerations (Frick, 2024). AI systems fall short of fully grasping human communication (Baidoo-Anu & Ansah, 2023; Marcus & Davis, 2020) as they cannot interpret emotional subtleties and human relationships' complexities. Moreover, AI does not possess the capacity for ethical discernment (Damasio, 2019; Swan, 2021), a domain in which humans evaluate values through culture, personal experience, and emotional factors. In modern society, recognizing the limitations of AI and understanding it as a technology shaped by human agency is essential

(Cukurova & Miao, 2024). Recent advancements in AI research, particularly the emphasis on human-centered AI, reveal a growing societal consensus that AI should support and augment human capabilities rather than replace them (Capel & Brereton, 2023; Garibay et al., 2023; Shneiderman et al., 2020). Humans must be central throughout the AI life-cycle—design, implementation, evaluation, operation, maintenance, decommissioning, and governance—to ensure high levels of oversight and control (Bond et al., 2019; Shneiderman, 2020). A *human consciousness* approach is necessary in AI design and use, aligning AI with human values, protecting human agency, and embedding social and ethical responsibility. Critical reflection on the extent to which AI can replace human cognition and decision-making, along with active participation in creating an AI-integrated society, is imperative. This emphasizes the need for thoughtful integration of AI in education across all life stages and academic disciplines. This requirement is predicated on the understanding that human consciousness represents the quintessential human attribute in a future society intertwined with AI, denoting the pinnacle of AI literacy essential for navigating a life augmented by AI technologies.

This study has several limitations that should be acknowledged. First, by focusing solely on peer-reviewed journal articles, we may have inadvertently excluded valuable insights from grey literature such as conference proceedings, accreditation documents, and policy papers. This selective approach, while ensuring academic rigour, potentially limits the breadth of our analysis. Second, the relatively small number of studies synthesized in this review may pose challenges to the generalizability of our findings. However, this limitation is balanced by the depth of analysis and the valuable integration of diverse perspectives on AI literacy within specific educational contexts, which aligns with our study's primary objective. Third, our exclusion of studies related to the education and training of AI developers or programmers was a deliberate choice based on our focus on AI literacy as a fundamental skill. While this decision allowed us to maintain a clear distinction between basic literacy and specialized technical competencies, it inevitably narrows the scope of our research. Literacy, as a basic human skill encompassing the ability to read, write, and comprehend (Berry et al., 1997; Tavdgiridze, 2016), forms the foundation of our study. This focus on fundamental skills differentiates our research from studies on specialized AI competencies required for AI professionals in the field. Moreover, the exclusion of non-English studies likely resulted in the omission of diverse perspectives and approaches from researchers worldwide, narrowing our understanding of global trends in AI-based education. Future research should address these limitations by expanding the scope to include grey literature, increasing the sample size of studies, and potentially incorporating perspectives from AI professional training. Such expansions would contribute to a more comprehensive understanding of AI literacy across various contexts and skill levels. Furthermore, we anticipate that this framework will serve as a cornerstone for forthcoming empirical inquiries, facilitating the refinement of the proposed framework. For example, it is imperative to develop measurement tools based on the competency framework derived from this study. Such tools would enable the assessment of learners' current states, and facilitate the application of tailored educational programs. This direction promises to enhance the effectiveness of AI literacy education and contributes to the personalized advancement of learner competencies in navigating AI environments.

CONCLUSION

The evolution of technology towards increased user-friendliness and intuitiveness marks a significant shift from an era requiring substantial training for technological proficiency to one in which such extensive preparation is largely unnecessary. This leap in the usability of technology, from the intensive training once necessary for PC mastery to the intuitive engagement with advanced devices and generative AI, shifts the focus of literacy related to that technology, in this study, AI. In other words, what matters is not the use of AI but the integration of AI into work and life. Therefore, AI education should progress from teaching how to use AI to teaching how to integrate it critically, strategically, responsibly, and ethically into our lives and professions, underpinned by a deeper understanding of the technology and underlying algorithms and their broader societal impacts and implications. This study demonstrates that AI and AI education exemplify this transition. We hope that this study not only contributes to educational practices and research but also inspires a proactive approach to embracing AI, ensuring that it serves as a beneficial and sustainable extension of human capability.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request to the corresponding author.

ETHICS STATEMENT

As this study is a systematic review and does not involve human subjects, ethics approval was not required.

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APPENDIX

TABLE A1 Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Publication date	January 1, 2012–October 1, 2024	
Language	English	Not English
Publication	Peer-reviewed journal publication	Non-peer-reviewed publication
Type of study	Original research (quantitative, qualitative, and review studies)	Commentary, letter, editorial, and book chapter
Context	K-12 education, higher education, teacher education, civic education, vocational education	Related to AI/computer science expert training such as that for AI developers
Discipline	Education	Studies from disciplines other than education, such as engineering, math, philosophy, or journalism
Study focus	Studies in the field of education on AI literacy (education *for* AI)	Studies that mention AI literacy a few times superficially without direct discussion and studies on use of AI for education (education *with* AI)

TABLE A2 Data extracted from reviewed studies.

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
1	Alexandre et al. (2021)	Mixed (survey, observation)	Citizens over 15 who have taken an AI MOOC ($n = 13{,}000$)	• Understanding of what AI means, the principles of machine learning, the crucial role played by data, and the social issues of AI	• Having a critical and creative perspective on AI	General public education
2	Ali et al. (2021)	Mixed (survey, interview)	Middle and high school students aged 10–15 years. ($n = 38$)	• Identifying fake media generated by AI and understanding the dangers of misuse of AI	• Having critical thinking about AI-generated media • Understanding the impact of AI-generated media on real-world policy	K-12 education
3	Cantú-Ortiz et al. (2020)	Qualitative (case study)	Participants in AI academic programs	• The ability to improve existing technologies and generate new ones such as AI	• Having problem-solving skills through collaborative activities using AI	Higher education (engineering)
4	Cardon et al. (2023)	Quantitative (survey)	Business communication instructors ($n = 323$)	• AI-assisted writing competency involving application, authenticity, accountability, and agency to communicate effectively	• Understanding AI tools and aligning AI tools with the task • Focusing on genuine communication and inserting human element • Taking responsibility for content reliability and ensuring equity and fairness • Retaining decision-making rights and avoiding long-term overreliance on AI	Workforce education (communication)

(Continues)

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
5	Chiu et al (2020)	Review (narrative)		• An individual's ability to clearly explain how AI technologies work and impact society, as well as to use them in an ethical and responsible manner and to effectively communicate and collaborate with them in any setting	• Knowing how AI technologies work • Understanding how AI technologies impact society • Using AI technologies in an ethical and responsible manner • Communicating and collaborating effectively with AI technologies in any setting • Having confidence and ability to self-reflect understanding for further learning	K-12 education
6	Eguchi et al. (2021)	Review (narrative)	–	• Understanding and utilizing sociotechnical system of AI and AI-enhanced technologies in the future	• Knowing the basic mechanics of AI • Understanding that all technical systems are sociotechnical systems • Recognizing sociotechnical system • Applying AI knowledge and applications • Considering the impact of technology on the world	K-12 education
7	Henry et al. (2021)	Mixed (Interview, observation)	Trainee teachers ($n=60$) Middle school students ($n=70$) Elementary pupils ($n=12$)	• Understanding the techniques and concepts underlying AI products and services	• Understanding and defining AI appropriately	K-12 education

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
8	Hilao-Valencia and Ortega-Dela Cruz (2023)	Mixed (survey, interview)	Teacher education students in college ($n=11$)	Not defined	• Communicating with AI and peers • Creating ideas using AI • Breaking down and organizing ideas using AI	Higher education (teacher education)
9	Huang (2021)	Quantitative	Senior middle school students ($n=978$)	• Experience AI achievement, understand the AI principle, and implement the application of AI	• Knowing basic AI concepts and cultural context • Cooperating/interacting with team members and tools • Analysing, exploring, practicing, raising questions about problems, and creating learning outputs with AI application	K-12 education
10	Karaca et al. (2021)	Quantitative	Medical students ($n=897$)	• The healthcare provider's preparedness state in terms of knowledge, skills, and attitude to utilize healthcare AI applications	• Having knowledge about medical AI applications • Choosing appropriate medical AI applications • Explaining limitations, strengths, and weaknesses related to medical AI • Complying with legal and ethical norms and regulations	Higher education (medicine)

(Continues)

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
11	Kim et al. (2021)	Review (narrative)	–	• Problem-solving skills using learner's understanding of AI technology and supporting students as a facilitator • Being familiar with technologies and selecting suitable software and platforms • Attention to AI's ethical and social issues	• Facilitating PBL using AI • Knowing the fundamentals of AI • Having basic knowledge of computer science • Knowing AI ethics • Using ICT tools and software • Constructing a programming environment • Using online AI education platforms • Providing feedback and encouraging peer review of project outcomes	Teacher education
12	Kim and Kwon (2023)	Mixed (survey, interview)	Elementary school teachers ($n=67$)	Not defined	• Content, pedagogical, and technological knowledge of AI • Technical content knowledge of AI • Technical pedagogical knowledge of AI • Pedagogical content knowledge of AI	Teacher education
13	Kong et al. (2023)	Quantitative (survey)	Students aged 15–18 years ($n=141$)	• Understanding AI concepts to solve real-world problems	• Developing AI concepts • Empowering individuals to confidently participate in the larger digital society • Concerning ethics of AI technology	K-12 education

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
14	Laupichler, Aster, et al. (2022)	Quantitative (survey)	Medical students ($n=24$)	• The healthcare provider's preparedness state in terms of knowledge, skills, and attitude to utilize healthcare AI applications during delivering prevention, diagnosis, treatment, and rehabilitation services in combination with own professional knowledge	• Analysing the data obtained by AI in healthcare • Understanding how AI applications in healthcare offer solutions to specific problems • Being able to explain the AI applications used in healthcare services to patients • Using health data in accordance with legal and ethical norms	Workforce education (healthcare)
15	Malerbi et al. (2023)	Review (narrative)		• Understanding, interpreting, and meaningfully criticizing the outputs of AI models	• Acquiring basic knowledge of machine learning and neural networks • Knowing how to access and analyse datasets and evaluating them critically • Understanding and avoiding biases and dangerous AI results, ensuring safe implementation • Integrating clinical workflows, bias control, and human–machine interaction in clinical settings • Understanding the legal and ethical aspects of digital healthcare and the impact of AI adoption • Building trust and refuting false beliefs related to AI results	Workforce education (healthcare)
16	Mamyrbayev et al. (2019)	Quantitative	–	Not defined	• Having critical thinking • Using and adapting information quickly in any situation • Finding and explaining information	Higher education (information technology)

(Continues)

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
17	Ng et al. (2021)	Review (systematic)		• Understanding AI concepts and ethical concerns to use AI responsibly	• Knowing and understanding AI • Using and applying AI • Evaluating and creating AI • Concerning ethical issues about AI	K-12 education Higher education Civic education Teacher education
18	Ng et al. (2023)	Mixed (scale development, score validation)	Secondary school students ($n=363$)	Not defined	• Applying, evaluating and creating AI • Knowing and understanding AI • Having confidence and self-efficacy in using AI • Having interest and motivation to learn AI • Collaborate with others in AI environment • AI ethics	K-12 education
19	Sanusi et al. (2022)	Quantitative (survey)	K-12 students ($n=605$)	Not defined	• Knowledge competence: skill competence and cultural competence • Learning competence: cognitive competence and self-learning competence • Team competence: teamwork competence and human–tool collaboration competence	K-12 education
20	Southworth et al. (2023)	Qualitative (curriculum design)	–	• The ability to understand, use, evaluate, and ethically navigate AI	• Enabling AI • Knowing and understanding AI • Using and applying AI • Evaluating and creating AI • Concerning AI ethics	Higher education

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
21	Swan (2021)	Quantitative	Nursing students (n = 675)	Not defined	• Having common knowledge of AI and knowing how it can improve processes and outcomes on the job site • Having skills to support information management platforms • Having statistical knowledge/ skills related to analytics, data management, and algorithm awareness • Having leadership for problem solving with AI technologies • Having knowledge for communication and collaboration with individuals using AI	Higher education (nursing)
22	Wang and Lester (2023)	Review (narrative)		• The ability to readily engage with AI by leveraging AI tools, systems, and frameworks to effectively and ethically solve problems in a wide range of sociocultural contexts	• Understanding AI capabilities • Utilizing AI for problem solving • Applying AI in sociocultural contexts	K-12 education
23	Wartman and Combs (2019)	Review (narrative)	–	Not defined	• Having skills to manage knowledge rather than retaining information • Collaborating with AI applications • Understanding AI and applying it meaningfully	Higher education (medical)

(Continues)

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
24	Wiljer and Hakim (2019)	Review (narrative)	–	• The capabilities required to navigate the complex world of AI • Appropriate and meaningful decision making through understanding and utilization of AI	• Understanding data governance principles • Knowing basic statistics and algorithmic decision making • Understanding data visualization and storytelling capabilities	Workforce education organization
25	Williams et al. (2023)	Qualitative (curriculum design)	78 students (aged 12.44 ± 1.32 years)	Not defined	• Having technical AI knowledge: understanding AI algorithms and applications • Ability to think critically about the implications of AI: foreseeing the ethics and social impact of AI systems • Ability to apply AI knowledge: creating collaborative human–AI systems	K-12 education
26	Yuan Tsai, & Chen (2024)	Mixed (scale development, score validation)		• The proficiency and capability to use AI products effectively and ethically	• Knowing what AI is and what AI can do • Understanding how AI works • Understanding how algorithms curate content and recognizing that algorithms make automated decisions • Understanding Human-AI interactions • Having response efficacy in using AI • Understanding ethical considerations over AI use	K-12 education Higher education Teacher education General public education

TABLE A2 (Continued)

	Study	Method	Participants	Definitions of AI literacy	Competencies comprising AI literacy	Target
27	Zanca et al. (2021)	Review (narrative)	–	Not defined	• Analysing medical images using AI • Using AI applications • Utilizing big data information • Identifying quality, regulatory, and ethical issues with AI	Workforce education (medical physics)
28	Zhang et al. (2023)	Quantitative (survey)	Middle school students ($n=25$)	• Understanding technical concepts; ethical and societal implications; and AI-related careers to become AI literate citizens	• Recognizing AI • Identifying features of AI technology • Understanding logic systems • Knowing general machine learning concepts • Understanding supervised learning, neural networks, and general adversarial networks • Being aware of AI-related careers and their futures • Understanding the ethical and societal implications of AI	K-12 education
29	Zhao et al. (2022)	Quantitative (survey)	Teachers of primary, secondary, and higher education ($n=1013$)	• Critical skills that individuals need to learn and live in the digital world with the help of AI-based technologies	• Understanding the basic concepts, knowledge, and instructions about using AI in teaching • Knowing when and how to use AI in the class • Evaluating AI technologies judiciously and commenting on the application of AI in teaching	Teacher education

TABLE A3 Definitions of AI literacy comparing to digital literacy.

Digital literacy framework (Law et al., 2018)		Competencies for AI literacy
0. Devices and Software Operations	→	**AI Devices and Software/Application/ Platform Operations**
To identify and use hardware tools and technologies. To identify data, information, and digital content needed to operate software tools and technologies		Build an environment for running AI processes and know how to use AI devices and software/ applications/platforms
1. Information and Data Literacy	→	**Data and Algorithm Literacy**
To articulate information needs and to locate and retrieve digital data, information, and content. To judge the relevance of the source and its content. To store, manage, and organize digital data, information and content		Understand the basic use of data and information, recognize how algorithms work, and apply the knowledge for using AI software/ applications/platforms
2. Communication and Collaboration	→	**Communication and Collaboration**
To interact, communicate, and collaborate through digital technologies while being aware of cultural and generational diversity. To participate in society through public and private digital services and participatory citizenship. To manage one's digital identity and reputation		Use AI technology to promote human-to-human and human-to-tool communication and collaboration skills
3. Digital Content Creation	→	**AI content creation**
To create and edit digital content. To improve and integrate information and content into an existing body of knowledge while understanding how copyright and licences are to be applied. To know how to give understandable instructions for a computer system		Use AI-based software to create content or run programming tools (Scratch, Python, etc.) to develop creative outcomes
4. Safety	→	**AI ethics**
To protect devices, content, personal data, and privacy in digital environments. To protect physical and psychological health and to be aware of digital technologies for social well-being and social inclusion. To be aware of the environmental impact of digital technologies and their use		Increase awareness of opportunities and threats of AI while considering AI ethics and social issues
5. Problem Solving	→	**Problem Solving**
To identify needs and problems and to resolve conceptual problems and problem situations in digital environments. To use digital tools to innovate processes and products. To keep up to date with the digital evolution		Know what the problems are, select appropriate AI software, and make the best decisions through problem solving based on AI technology
6. Career-related Competences	→	**Career-related Competences**
To operate specialized digital technologies and to understand, analyse, and evaluate specialized data, information, and digital content for a particular field		Operate specialized AI technologies and understand, analyse, and evaluate specialized data, information, and AI-based content for a particular field
		Affective competence
		Have confidence and self-efficacy to develop interest and motivation in AI technologies, along with the ability for self-reflection in learning AI

TABLE A4 Competencies of AI literacy compared to ones of digital literacy.

Digital literacy framework (Law et al., 2018)	Competencies for AI literacy
0. Devices and software operations	**1. AI device and software/application/platform operations**
0.1 Physical operations of digital devices	1.1 Building an optimal environment for AI processes
0.2 Software operations in digital devices	1.2 Knowing how to operate AI software/applications/platforms
1. Information and data literacy	**2. Data and algorithm literacy**
1.1 Browsing, searching and filtering data, information and digital content	2.1 Understanding data and information
1.2 Evaluating data, information, and digital content	
1.3 Managing data, information, and digital content	
	2.2 Knowing and applying knowledge about AI/algorithms
5. Problem solving	**3. Problem solving**
5.2 Identifying needs and technological responses	3.1 Identifying and defining problems that should be solved using AI technology
5.4 Identifying digital competence gaps	
5.3 Creatively using digital technologies	3.2 Applying problem-solving process using AI technology
5.5 Computational thinking	
5.1 Solving technical problems	
	3.3 Data-based decision making
2. Communication and collaboration	**4. Communication and collaboration**
2.1 Interacting through digital technologies	4.1 Human-to-tool collaboration through AI technology
2.2 Sharing through digital technologies	4.2 Human-to-human communication with AI technology
2.3 Engaging in citizenship through digital technologies	
2.4 Collaborating through digital technologies	
2.5 Netiquette	
2.6 Managing digital identity	
4. Safety	**5. AI ethics**
4.2 Protecting personal data and privacy	5.1 Being aware of the opportunities and threats of AI 5.2 Knowing and being concerned about AI ethics
4.3 Protecting health and well-being	
4.1 Protecting devices	
4.4 Protecting the environment	
6. Career-related competences	**6. Career-related competences**
6.1 Operating specialized digital technologies for a particular field	6.1 Operating specialized AI technologies for a particular field
6.2 Interpreting and manipulating data, information, and digital content for a particular field	6.2 Interpreting and manipulating data, information, and AI content for a particular field
	6.3 Verifying AI results for accuracy and making rational decisions based on validated data

(Continues)

TABLE A4 (Continued)

Digital literacy framework (Law et al., 2018)	Competencies for AI literacy
3. Digital content creation	**7. AI content creation**
3.1 Developing digital content	7.1 Creating content utilizing AI tools
3.2 Integrating and re-elaborating digital content	
3.3 Copyright and licences	
3.4 Programming	7.2 Programming
	8. Affective competences
	8.1 Having confidence and self-efficacy to promote interest and motivation in using AI technologies
	8.2 Having a self-reflective mindset to continuously assess one's understanding of AI, identify one's level of AI literacy, and recognize areas that further learning

TABLE A5 Comparison of existing and proposed frameworks.

Target learner		This study K-16, teachers, organizations, workforce, general public	Long and Magerko (2020) K-12	Ng et al. (2021) K-16, teachers, general public	Ng et al. (2023) K-12	UNESCO (Cukurova & Miao, 2024) Teachers
Competency						
1. AI device and software/application/platform operations	1.1 Building an optimal environment for AI process	●				●
	1.2 Knowing how to operate AI software/application/platform	●		●	●	●
2. Data and algorithm literacy	2.1 Understanding data and information	●	●			●
	2.2 Knowing and applying knowledge about AI/algorithms	●	●	●	●	●
3. Problem solving	3.1 Identifying and defining problems that should be solved using AI technology	●	●			
	3.2 Applying problem-solving processes using AI technology	●	●		●	
	3.3 Data-based decision making	●	●	●	●	
4. Communication and collaboration	4.1 Human-to-tool (AI) collaboration through AI technology	●	●		●	●
	4.2 Human-to-human communication with AI technology	●	●		●	●

(Continues)

TABLE A5 (Continued)

Target learner		This study K-16, teachers, organizations, workforce, general public	Long and Magerko (2020) K-12	Ng et al. (2021) K-16, teachers, general public	Ng et al. (2023) K-12	UNESCO (Cukurova & Miao, 2024) Teachers
5. AI ethics	5.1 Being aware of the opportunities and threats of AI	●	●		●	●
	5.2 Knowing and being concerned about AI ethics	●	●	●	●	●
6. Career-related competences	6.1 Operating specialized AI technologies for a particular field	●				●
	6.2 Interpreting and manipulating data, information, and AI content for a particular field	●				●
	6.3 Verifying AI results for accuracy and making rational decisions based on validated data	●				
7. AI content creation	7.1 Creating AI content using AI tools	●	●	●	●	●
	7.2 Programming	●	●			
8. Affective competences	8.1 Confidence and self-efficacy in using AI	●			●	
	8.2 Self-reflective mindset towards AI	●				
* Human-centered mindset	* The values and attitudinal orientation towards human–AI interactions					●

*The 'Human-centered mindset' competency is not included in the framework as the UNESCO (Cukurova & Miao, 2024) document is classified as grey literature.